shows two things: the natural seasonal respiration of Earth and the year-on-year rise in atmospheric $CO_2$. Atmospheric $CO_2$ is significant because carbon dioxide is the most important of the greenhouse gases, which trap warmth in Earth's atmosphere. More molecules of $CO_2$ and other greenhouse gases cause more heat to be trapped, leading to an overall increase in temperature and global climate change.

**"We were witnessing for the first time nature's withdrawing $CO_2$ from the air for plant growth during summer and returning it each succeeding winter."**

**Charles Keeling**

## Measuring $CO_2$ levels

Since the start of the Industrial Revolution in the late 1700s, human activity has produced increasing emissions of $CO_2$. This is largely due to burning fossil fuels, while forest clearance for agriculture and development has resulted in less vegetation absorbing $CO_2$ through photosynthesis. Many scientists once believed that excess $CO_2$ would be absorbed by the oceans. Others disagreed, but there was little hard evidence either way.

Charles Keeling was not the first to propose a link between atmospheric warming and $CO_2$ emissions. Others had measured $CO_2$ levels but had produced only "snapshots" in time rather than a long-term dataset. Keeling knew that a long study was needed to prove the link. In 1956, he took up a post at the Scripps Institution of Oceanography in San Diego, California, and obtained funds to establish $CO_2$ monitoring stations at remote locations 9,843 ft (3,000 m) up on Mauna Loa, Hawaii, and at the South Pole. By 1960, Keeling was sure that he had a long-enough series of records to detect a year-on-year increase.